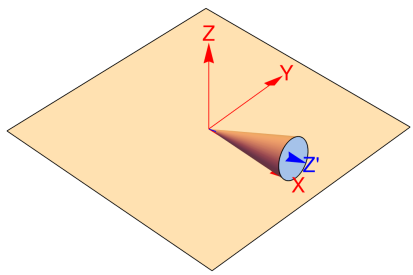
Homework 2

(***Deadline***: May 02, 2026)

1. (***2 points***)



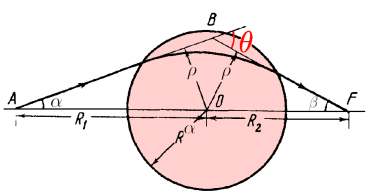
A homogeneous cone of mass m, radius R and length $\ell \gg R$ rolls over the horizontal surface, as shown in the figure. The tip (vertex) of the cone does not move, but the cone can rotate freely around it.

(a) (***1 point***) Write out the Lagrangian for the system. Identify its symmetries, cyclic variables and integrals of motion that may exist in this system.

(b) (***1 point***) Write out the Euler-Lagrange equations and integrate them. You may use integrals of motion to find the trajectories.

***Help*:** The motion of the cone may be represented as a combination of rotation around Z' (symmetry axis of the cone) and rotation of that axis around Z (vertical axis, normal to the plane). The angle between axes Z and Z' remains constant. Try to use this information to parametrize the rotation in terms of the Euler angles. Since the cone rolls (does not slide), these rotation angles may be related.

2. (***2 points***)



The electrons are emitted isotropically at the point A and are focused in the point F after scattering in the central potential (colored area).

The electrons are emitted isotropically with constant energy E at the point A. We need to create the central potential $U(r)$ that will focus them at the point F located at some distance. The center of the field (point O) is located on the line connecting the points A and F, at distances R_1 and R_2 from them, respectively. For the sake of simplicity, You may assume that $R_1 = R_2 = R$, and that the potential vanishes at distances $r > R$ from the point O (the so-called "*Maxwell's fish-eye configuration*").

(a) (***1/2 point***) Write out the scattering angle $\theta = \alpha + \beta$ as a function of the impact parameter ρ.

(b) (***1 point***) Find the potential $U(r)$ which satisfies these properties.

(c) (***1/2 point***) Assume that we analyze scattering of a collinear beam of monochromatic particles in the field which You found earlier. Find the cross-section of elastic scattering $d\sigma/d\Omega$ for this case.

Help: We discussed inverse problems of scattering in Lecture 9.